EXPERIMENT 6

Implementation of PCA, SVD and LDA

Dataset Used

Iris Dataset — 150 samples, 4 features and 3 classes: setosa, versicolor and virginica.

Program

```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

# Load Iris Dataset
iris = load_iris()

X = iris.data
y = iris.target

print("========== DATASET ==========")
print("Dataset Shape:", X.shape)
print("Features:", iris.feature_names)
print("Classes:", iris.target_names)

# Standardize Data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("\nStandardized Data Shape:", X_scaled.shape)

# PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("\n========== PCA ==========")
print("Explained Variance Ratio:")
print(np.round(pca.explained_variance_ratio_, 4))

print("Total Variance Retained:")
print(round(pca.explained_variance_ratio_.sum(), 4))

print("Reduced Shape:")
print(X_pca.shape)

print("\nFirst 5 PCA Values:")
print(np.round(X_pca[:5], 4))

# SVD
U, S, VT = np.linalg.svd(X_scaled, full_matrices=False)

X_svd = U[:, :2] @ np.diag(S[:2])

print("\n========== SVD ==========")
print("Singular Values:")
print(np.round(S, 4))

print("Reduced Shape:")
print(X_svd.shape)

print("\nFirst 5 SVD Values:")
```

```
print(np.round(X_svd[:5], 4))

# LDA
lda = LinearDiscriminantAnalysis(n_components=2)
X_lda = lda.fit_transform(X_scaled, y)

print("\n========== LDA ==========")
print("Explained Variance Ratio:")
print(np.round(lda.explained_variance_ratio_, 4))

print("Reduced Shape:")
print(X_lda.shape)

print("\nFirst 5 LDA Values:")
print(np.round(X_lda[:5], 4))

# PCA Graph
plt.figure(figsize=(7, 5))
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - Iris Dataset")
plt.show()

# SVD Graph
plt.figure(figsize=(7, 5))
plt.scatter(X_svd[:, 0], X_svd[:, 1], c=y)
plt.xlabel("SVD Component 1")
plt.ylabel("SVD Component 2")
plt.title("SVD - Iris Dataset")
plt.show()

# LDA Graph
plt.figure(figsize=(7, 5))
plt.scatter(X_lda[:, 0], X_lda[:, 1], c=y)
plt.xlabel("Linear Discriminant 1")
plt.ylabel("Linear Discriminant 2")
plt.title("LDA - Iris Dataset")
plt.show()
```

Output

```
========== DATASET ==========
Dataset Shape: (150, 4)
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Classes: ['setosa' 'versicolor' 'virginica']

Standardized Data Shape: (150, 4)

========== PCA ==========
Explained Variance Ratio:
[0.7296 0.2285]
Total Variance Retained:
0.9581
Reduced Shape:
(150, 2)

First 5 PCA Values:
[[-2.2647  0.48  ]
 [-2.081  -0.6741]
 [-2.3642 -0.3419]
 [-2.2994 -0.5974]
 [-2.3898  0.6468]]

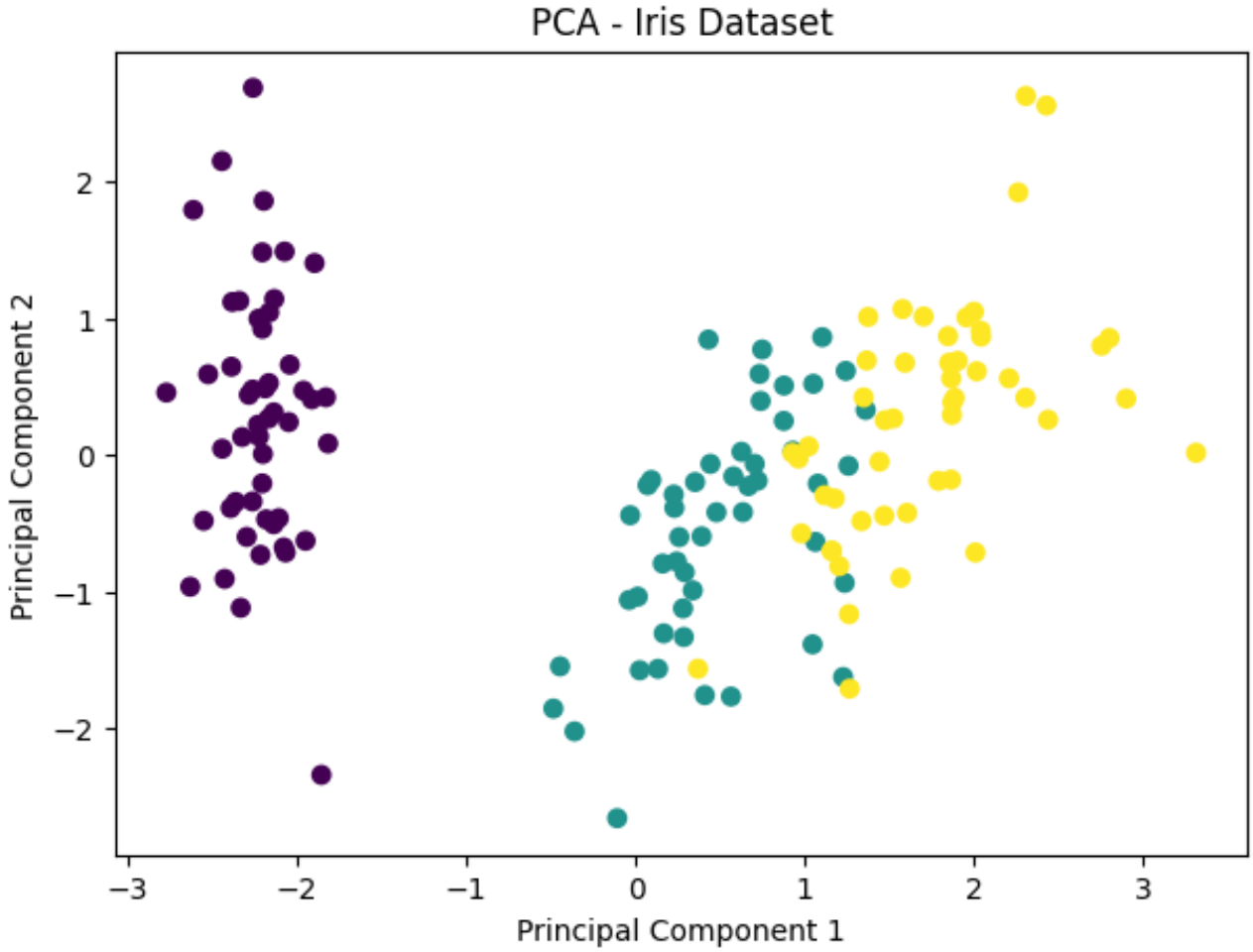
========== SVD ==========
Singular Values:
[20.9231 11.7092  4.6919  1.7627]
Reduced Shape:
(150, 2)

First 5 SVD Values:
[[-2.2647 -0.48  ]
 [-2.081   0.6741]
 [-2.3642  0.3419]
 [-2.2994  0.5974]
 [-2.3898  0.6468]]

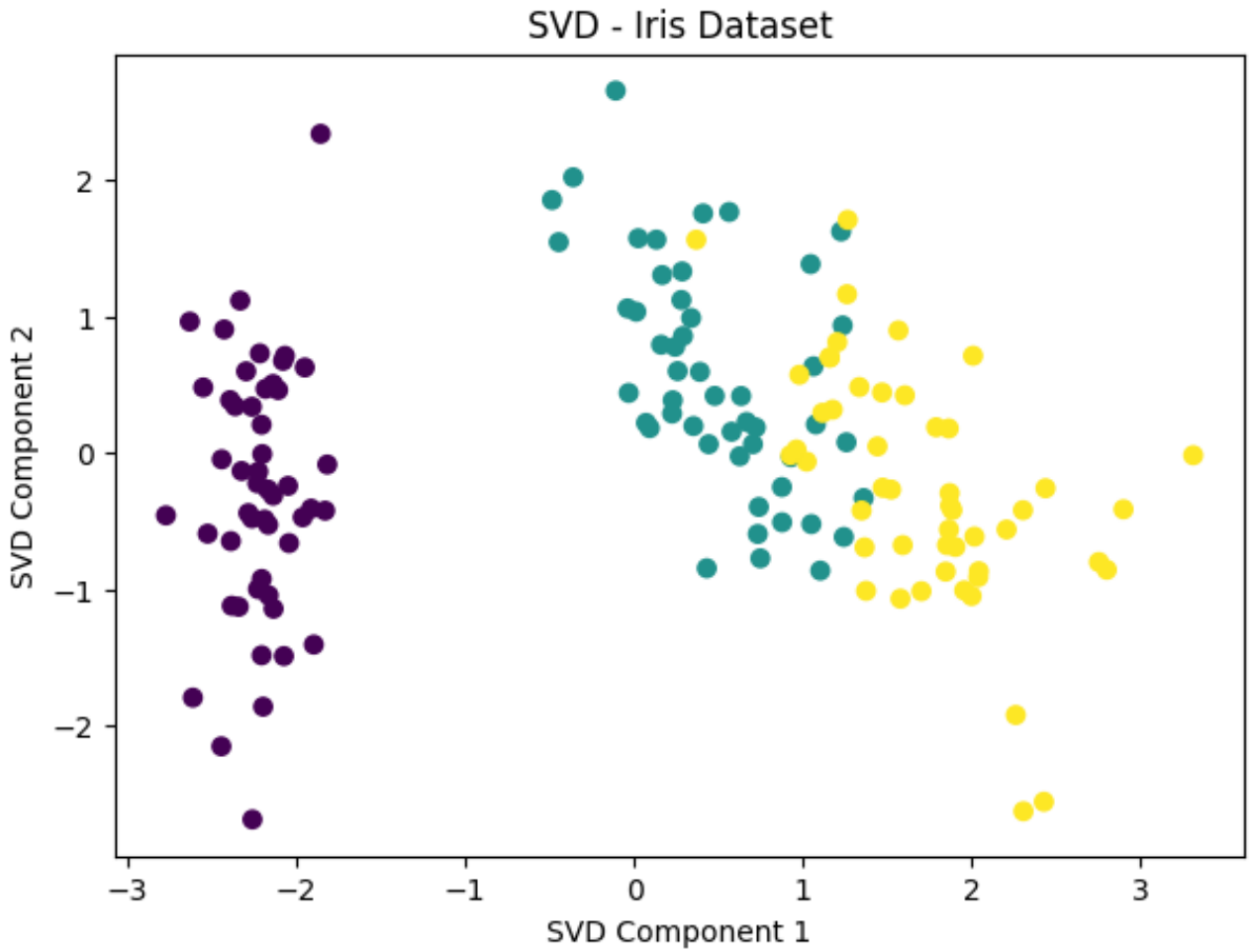
========== LDA ==========
Explained Variance Ratio:
[0.9912 0.0088]
Reduced Shape:
(150, 2)

First 5 LDA Values:
[[ 8.0618 -0.3004]
 [ 7.1287  0.7867]
 [ 7.4898  0.2654]
 [ 6.8132  0.6706]
 [ 8.1323 -0.5145]]
```

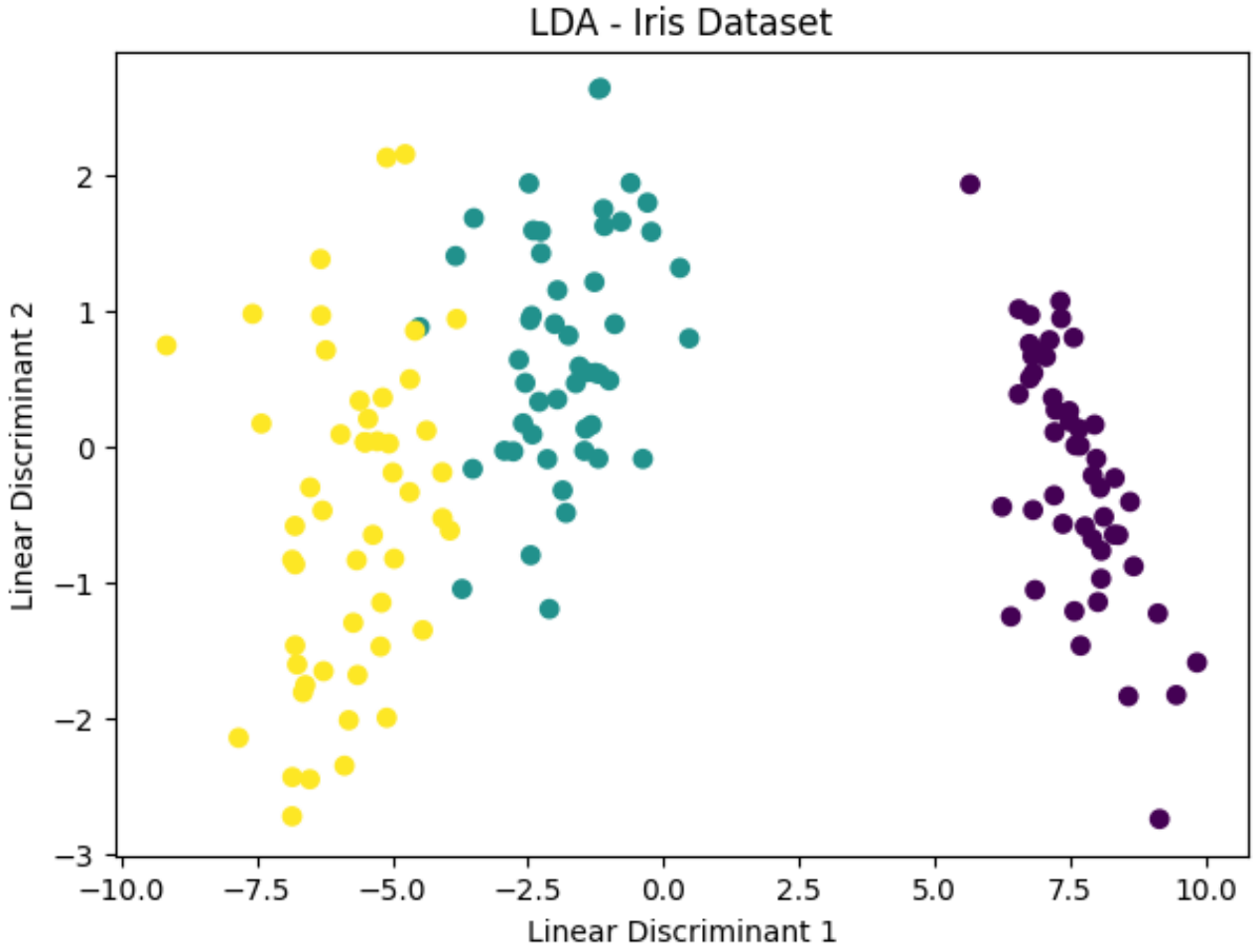
PCA Graph


PCA - Iris Dataset
Principal Component 2
Principal Component 1
2
1
0
−1
−2
−3
−2
−1
0
1
2
3

SVD Graph


SVD - Iris Dataset
SVD Component 2
SVD Component 1
2
1
0
−1
−2
−3
−2
−1
0
1
2
3

LDA Graph


LDA - Iris Dataset
Linear Discriminant 2
Linear Discriminant 1
2
1
0
−1
−2
−3
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0